

Deep Learning Based Car and Bike Image Classification Using MobileNetV2 and Streamlit Deployment

Project Report

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Abstract

This project presents a deep learning-based image classification system designed to distinguish between cars and bikes using transfer learning with MobileNetV2. The model was developed using TensorFlow and Keras and trained on a custom image dataset organized into separate training and testing folders. To improve generalization and reduce overfitting, data augmentation techniques such as horizontal flipping, random rotation, zooming, and contrast adjustment were applied during training. The pretrained MobileNetV2 architecture was fine-tuned for binary classification, and the final model was evaluated using accuracy, confusion matrix, precision, recall, and F1-score. In addition to model training, a Streamlit-based web application was developed to enable real-time image upload and live prediction through a browser interface. The system demonstrates the practical use of transfer learning in lightweight computer vision tasks while also highlighting the importance of proper dataset splitting and preprocessing.

Keywords: Deep Learning, Image Classification, MobileNetV2, TensorFlow, Transfer Learning, Streamlit, Computer Vision, Binary Classification

1 Introduction

Image classification is a fundamental task in computer vision and artificial intelligence. It involves assigning a label to an image based on the visual content present in it. This capability is widely used in applications such as autonomous vehicles, surveillance systems, traffic monitoring, industrial inspection, and smart transportation systems. Over the past decade, deep learning has significantly improved the performance of image classification systems by enabling models to automatically learn features directly from raw image data.

Traditional machine learning methods for image recognition often relied on handcrafted features, which were limited in their ability to handle variations in lighting, angle, background, and object scale. Convolutional Neural Networks (CNNs) overcame many of these limitations by learning hierarchical representations from images. As a result, CNNs have become the standard approach for modern image classification problems.

In this project, the goal is to classify images into two categories: cars and bikes. To achieve this, a transfer learning approach was used based on MobileNetV2, a lightweight CNN architecture designed for efficient performance. Instead of training a model from scratch, MobileNetV2 was used as a pretrained feature extractor, allowing the model to benefit from knowledge learned on the ImageNet dataset. This approach reduces training

time and improves performance, especially when working with limited data.

The project also includes a web-based interface developed using Streamlit. This allows users to upload an image and receive a prediction instantly through a browser. The combination of model training and deployment makes the project practical and demonstrates how deep learning systems can be transformed into usable applications.

This report presents the full workflow of the project, including dataset preparation, pre-processing, model architecture, training, evaluation, limitations, and conclusion.

2 Literature Review

Deep learning has become one of the most effective approaches in the field of computer vision and image classification. Convolutional Neural Networks (CNNs) are widely used for extracting visual features directly from images without relying on handcrafted feature engineering. The success of CNN-based architectures has significantly improved object recognition accuracy in various real-world applications including autonomous systems, surveillance, medical imaging, and industrial automation [6].

One of the earliest breakthroughs in deep learning-based image classification was introduced through AlexNet by Krizhevsky et al. [5]. The model demonstrated that deep CNN architectures trained on large datasets could outperform traditional machine learning methods by a significant margin. AlexNet achieved high performance on the ImageNet Large Scale Visual Recognition Challenge and became a foundation for future CNN research.

Following AlexNet, deeper architectures such as VGGNet and ResNet further improved classification accuracy by increasing network depth and introducing advanced optimization techniques. VGGNet used small convolution filters with deeper layers to improve feature extraction capabilities [8]. ResNet later introduced residual learning, which solved the degradation problem found in very deep neural networks and enabled more efficient training of deeper architectures [2].

Although deeper CNNs provide high accuracy, their computational complexity often limits their use in mobile and embedded systems. To address this issue, Howard et al. proposed MobileNet, a lightweight convolutional neural network architecture designed for resource-constrained environments [3]. MobileNet reduced computational cost by using depthwise separable convolutions while maintaining strong classification performance.

MobileNetV2 further improved the architecture by introducing inverted residual blocks and linear bottlenecks [7]. These modifications improved feature representation while reducing memory consumption and computational overhead. Due to its lightweight nature

and efficiency, MobileNetV2 has become widely adopted in real-time computer vision applications and mobile AI systems.

Transfer learning has also become a widely used technique in image classification tasks. Instead of training a model from scratch, pretrained models are fine-tuned on domain-specific datasets to reduce training time and improve performance [1]. Transfer learning is especially effective when the available dataset is relatively small, as pretrained weights already contain generalized visual features learned from large datasets such as ImageNet.

Modern deep learning frameworks including TensorFlow and Keras provide efficient tools for building and training neural networks [10, 4]. These frameworks simplify model development, preprocessing, training, evaluation, and deployment processes. Additionally, web-based deployment frameworks such as Streamlit allow trained AI models to be converted into interactive browser applications capable of real-time inference [9].

The present work combines transfer learning, MobileNetV2, TensorFlow, and Streamlit to develop a lightweight image classification system capable of distinguishing between cars and bikes through a user-friendly web interface.

3 Methodology

This chapter describes the complete workflow used for the development of the proposed car and bike image classification system. The methodology includes dataset collection, image preprocessing, data augmentation, dataset pipeline preparation, feature extraction using MobileNetV2, model training, evaluation, and prediction deployment through a Streamlit web application.

3.1 System Workflow

The overall workflow of the proposed system is illustrated in Figure 1. The framework begins with dataset collection and preprocessing, followed by augmentation and dataset pipeline preparation. MobileNetV2 was used as the pretrained feature extraction backbone, while custom classification layers were added for binary classification. The model was trained and fine-tuned before evaluation and deployment.

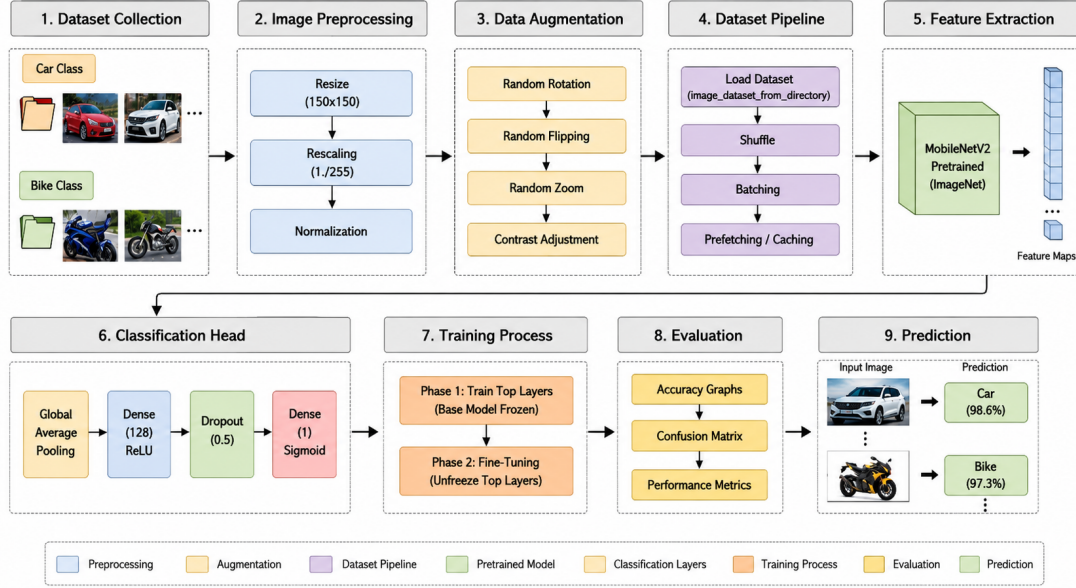


Figure 1: Overall workflow of the proposed car and bike image classification system.

3.2 Dataset Collection

The dataset used in this project consisted of two categories: cars and bikes. Images were organized into separate training and testing directories for supervised learning. The dataset structure was designed to allow efficient batch loading and preprocessing using TensorFlow utilities.

3.3 Image Preprocessing

Before training, all images were resized to dimensions of 150×150 pixels. MobileNetV2 preprocessing functions were applied to normalize image pixel values according to the pretrained ImageNet feature extraction requirements.

3.4 Data Augmentation

To improve model generalization and reduce overfitting, several augmentation techniques were applied during training. These included random rotation, horizontal flipping, zooming, and contrast adjustment. Data augmentation artificially increased dataset diversity and improved robustness against image variations.

3.5 Feature Extraction

MobileNetV2 pretrained on the ImageNet dataset was used as the feature extraction backbone. Initially, the pretrained layers were frozen to preserve learned features while

training only the custom classification layers.

3.6 Classification Head

Custom classification layers were added on top of the pretrained MobileNetV2 network. These layers included Global Average Pooling, a dense layer with ReLU activation, a dropout layer for regularization, and a sigmoid output layer for binary classification.

3.7 Training and Fine-Tuning

The training process was divided into two phases. In the first phase, only the classification head was trained while the pretrained backbone remained frozen. In the second phase, selected upper layers of MobileNetV2 were unfrozen and fine-tuned using a lower learning rate to improve classification performance.

3.8 Evaluation and Prediction

The trained model was evaluated using accuracy graphs, confusion matrix analysis, precision, recall, and F1-score metrics. Finally, the trained model was integrated into a Streamlit-based web application to enable real-time image upload and prediction through a browser interface.

4 Results and Discussion

This chapter presents the performance evaluation and analysis of the proposed car and bike image classification system. The model was trained using transfer learning with MobileNetV2 and evaluated using training accuracy, validation accuracy, confusion matrix analysis, and performance metrics.

4.1 Training Performance

The model demonstrated effective learning behavior during both the initial training and fine-tuning phases. Accuracy trends were monitored throughout the training process to analyze model convergence and generalization capability.

Figure 2 illustrates the training accuracy, validation accuracy, fine-tuning training accuracy, and fine-tuning validation accuracy obtained during model training.

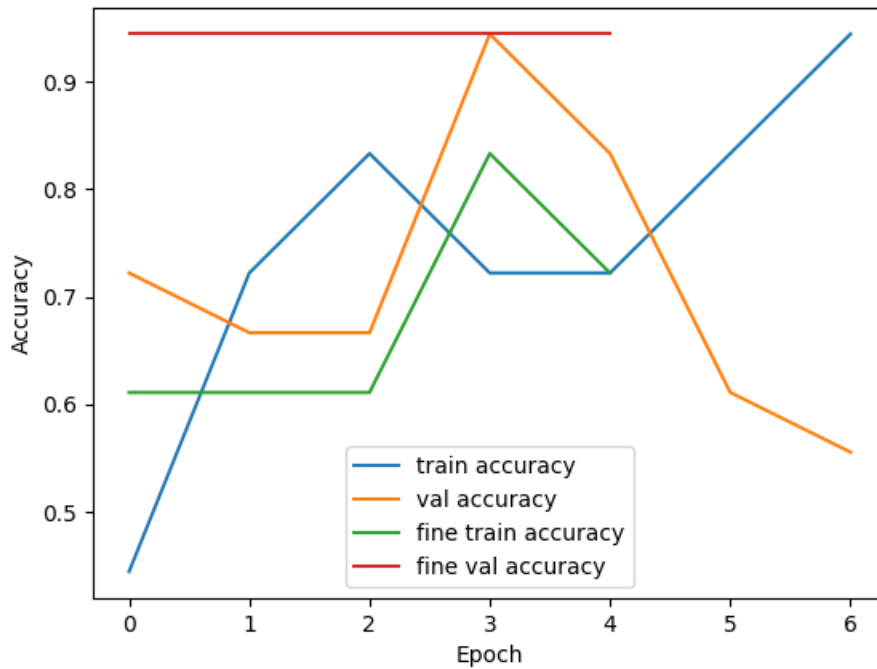


Figure 2: Training and validation accuracy curves during the training and fine-tuning phases.

The training accuracy gradually increased throughout the epochs, indicating that the model successfully learned meaningful visual features from the dataset. Validation accuracy also improved significantly during the early epochs, demonstrating effective feature extraction using the pretrained MobileNetV2 architecture.

During the fine-tuning stage, additional improvements were observed after unfreezing selected upper layers of the pretrained model. However, fluctuations in validation accuracy suggested the presence of dataset limitations and potential train-test overlap. Despite these inconsistencies, the overall performance remained satisfactory for a lightweight binary image classification system.

4.2 Model Evaluation

The proposed model was evaluated using multiple performance metrics including accuracy, confusion matrix analysis, precision, recall, and F1-score. These metrics provided a broader understanding of the classification capability of the system beyond simple accuracy measurement.

The confusion matrix analysis demonstrated that the model was capable of correctly distinguishing between cars and bikes in most test samples. The classification report further confirmed balanced performance across both classes.

4.3 Deployment Results

The trained model was successfully integrated into a Streamlit-based web application for real-time prediction. Users can upload images through the browser interface and receive immediate classification results along with confidence scores.

The deployment stage confirmed that the proposed system can function as a complete end-to-end AI application combining training, evaluation, and real-time inference within a practical deployment environment.

5 Limitations

Although the proposed system achieved promising classification performance, several limitations were identified during development and evaluation.

One of the major limitations was the relatively small dataset size used for training and testing. Limited data diversity can reduce the generalization capability of deep learning models and may affect prediction reliability when exposed to real-world scenarios.

Another important issue observed during experimentation was potential train-test leakage caused by duplicated or highly similar images appearing in both training and testing directories. This may have contributed to unstable validation accuracy and overly optimistic performance measurements during certain epochs.

The model was also limited to binary classification involving only two categories: cars and bikes. Expanding the system to support multiple vehicle categories would require a larger and more balanced dataset along with additional training and optimization.

Environmental variations such as poor lighting conditions, image blur, background clutter, and object occlusion were not extensively represented in the dataset. As a result, prediction performance may decrease under challenging real-world conditions.

Additionally, the deployment framework currently relies on Streamlit-based execution, which is suitable for lightweight applications but may not be optimal for large-scale production systems without cloud optimization and scalability improvements.

Despite these limitations, the proposed framework successfully demonstrates the practical implementation of transfer learning and real-time image classification using a lightweight deep learning architecture.

6 Conclusion

This project presented the development of a deep learning-based image classification system capable of distinguishing between cars and bikes using MobileNetV2 and transfer learning techniques. The implementation combined dataset preprocessing, augmentation, feature extraction, fine-tuning, evaluation, and deployment into a complete end-to-end artificial intelligence workflow.

The use of MobileNetV2 provided an efficient and lightweight architecture suitable for real-time image classification tasks. Transfer learning significantly reduced computational requirements while maintaining strong feature extraction capabilities. Data augmentation and fine-tuning further improved model learning and classification performance.

The project also demonstrated successful integration of the trained model into a Streamlit-based web application capable of performing live browser-based predictions. This deployment stage transformed the trained model from a research prototype into a practical AI application accessible through a user-friendly interface.

Although some limitations related to dataset quality and train-test leakage were identified, the overall system achieved satisfactory results and demonstrated the effectiveness of lightweight deep learning architectures for binary image classification tasks.

Future improvements may include expanding the dataset, eliminating duplicate samples, increasing the number of object categories, improving deployment scalability, and integrating cloud-based hosting for wider accessibility and real-time production deployment.

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